

TrustEpisode: Auditable Evidence Contracts for AI-Assisted EDR/NDR Investigation

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Abstract

AI-assisted security investigation requires more than grounded prompts: the underlying case state must remain reproducible, versioned, and independently auditable under delayed delivery and incomplete telemetry. We present TRUSTEPISODE, a canonicalized, single-writer reconstruction and audit contract for EDR/NDR evidence. Episode Synthesis alone owns membership; revisions and terminal audit outcomes are append-only; authenticated source health is separated from event silence; unsupported probabilities remain JSON null; and downstream AI consumes version-scoped records without authority over case facts.

On 70 primary malicious formation runs (40 standalone benign controls are execution controls, not detection trials), the sealed cohort shows a descriptive mean coverage increase from 0.807 to 0.821 versus a legacy fixed-bin reference. Fair sliding builders do not establish formation superiority: EB4 ties EB2 on all action-coverage pairs (0/70/0 W/T/L) and trails EB1 on one concurrent-benign M7 run (0/69/1; mean $\Delta = -0.014$, 95% CI $[-0.043, 0.000]$). The cohort therefore mainly exercises sliding sessionization rather than relation-specific gains.

Canonical replay is byte-identical on 60 M1–M6 runs under re-canonicalization; raw arrival order is not robust. 18/18 fixtures exercise all seven AssemblyFailureRecord codes; an A–G taxonomy localizes M2’s gap to token mapping; and 8/8 downstream cards reject six invalid or unsealed mutations while accepting a resealed allowed-field control. Once-run Qwen2.5-7B and locked Gemma3-1B audits yield 9/20 and 0/20 fully exact responses. Rather than making an untrusted model inherently accurate, TRUSTEPISODE makes its evidence use falsifiable against immutable, version-scoped case records.

CCS Concepts

• Security and privacy → Intrusion/anomaly detection and malware mitigation; • Computing methodologies → Artificial intelligence.

Keywords

AI-assisted security operations, attack episode reconstruction, EDR, NDR, replayable audit, evidence provenance

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1 Introduction

AI-assisted SOC systems increasingly use language-model agents to gather tool outputs, correlate alerts, and draft incident narratives [5, 21, 25]. Without a stable evidence substrate, those narratives are hard to audit: arrival order can change episode membership, late evidence can rewrite prior history, collector silence can be mistaken for benign absence, and a model can fill missing steps from conversational context until a generated assertion becomes indistinguishable from case fact.

Security operations and digital-forensic investigations therefore need more than alert clustering. Multi-step attacks appear as dispersed EDR and NDR observations that become material when linked by time, entity, session, and process relations [1, 11, 15]. Investigators need a contract that fixes who owns episode membership, how late evidence is admitted, how collector health differs from “no detection,” and how every revision leaves an independently verifiable terminal outcome [3, 4, 16].

Prior alert-correlation, provenance, and forensic systems optimize complementary research units: scenario graphs and alert chains [12]; suspicious provenance subgraphs [6, 8–10, 20]; and evidence-grounded agent interfaces that cite tool outputs [21, 25]. They leave open the need for fixed membership ownership, immutable revision history, explicit missingness, and replayable audit semantics when telemetry is incomplete. TRUSTEPISODE addresses that lifecycle contract: reconstruction deterministic over canonicalized, version-bound inputs, plus a typed read-only boundary for downstream AI. The model remains outside the trusted computing base: it may summarize versioned AuditRecords but cannot own membership, infer source health, overwrite history, or invent unsupported probabilities. Our AISeC claim is architectural containment of evidence use, not broad LLM effectiveness. Held-out Qwen and locked Gemma audits measure extraction, citation, and abstention as boundary tests, not model ranking.

Research questions.

- RQ1** Can TrustEpisode reproduce byte-identical membership, revision history, late-evidence records, and digests on 60 sealed malicious runs?
- RQ2** Do open, finalized, late-successor, closed, beyond-horizon, duplicate, and all seven AssemblyFailureRecord codes yield explicit immutable audit outcomes without rewriting earlier history?
- RQ3** When a required action is absent from a reconstructed episode, can the pipeline localize loss across the closed A–G

taxonomy, and is that taxonomy instantiated on sealed live evidence?

RQ4 Which relation, hub, merge, and late-revision mechanisms does the live M1–M7 cohort exercise, and which require targeted conformance cards?

RQ5 What descriptive coverage change accompanies replacing fixed bins with sliding sessionization on the sealed cohort, and do fair relation-aware builders show additional measurable benefit?

RQ6 Do closed schemas and digest bindings fail closed under invalid or unsealed mutations; what remains possible after a schema-valid reseal; and how do two local models follow the resulting read-only extraction, citation, and abstention interface?

These questions evaluate the evidence substrate. The deployed interface trace checks end-to-end materialization but is not an effectiveness trial; its displayed operational confidence is not the contract’s unevaluated calibrated probability.

Contributions.

- (1) **Contract.** A canonicalized single-writer evidence lifecycle with immutable revisions, late-evidence handling, source-health separation, canonical references/digests, and exactly one terminal AssemblyOutcome per revision (Sections 3–3.6).
- (2) **Evidence-class evaluation.** A sealed evaluation separating live formation, offline replay, synthetic conformance, and local-model audits: 70 malicious M1–M7 runs, 40 standalone benign execution controls, fair EB0–EB4 baselines, and a 60-run determinism matrix (Section 4).
- (3) **Diagnostics and AI boundary.** A closed A–G evidence-loss taxonomy instantiated on M2, plus downstream mutation cards that reject six invalid or unsealed mutations and accept a resealed allowed-field control, exposing why writer authorization remains external (Sections 4.10–4.13).
- (4) **Reproducibility and deployment trace.** An executable anonymous companion and a read-only deployed case trace showing that source-separated Suricata/Wazuh evidence and ATT&CK references survive into an analyst-facing view (Sections 4.6, 4.17).

2 Related Work

2.1 Alert Correlation and Attack Reconstruction

Classical alert correlation reconstructs multi-step scenarios from intrusion alerts by temporal and causal linking [12]. HOLMES and UNICORN recover APT context from system graphs [8, 10]; KAIROS, NODLINK, and CAPTAIN extend this line [6, 9, 20]. These systems motivate episode construction. Our research unit is instead the lifecycle contract governing mixed-EDR/NDR membership, late-data revisions, and terminal audit outcomes; detection and provenance context are complementary inputs, not effectiveness baselines.

2.2 Digital Forensic Event Reconstruction

Timeline-based event reconstruction is a core digital-forensic technique for inferring past activity from heterogeneous artifacts [1, 4]. Recent work formalizes timestamp interpretation via time anchors and clock-skew anomalies [16] and surveys challenges beyond mere data volume [1]. Process models and filesystem analysis emphasize documenting what was examined, what was unavailable, and how conclusions depend on incomplete traces [3, 11]. TrustEpisode imports these audit requirements into *online* EDR/NDR episode reconstruction: immutable revisions, explicit late stores, and digestable terminal outcomes under delayed delivery.

2.3 Reproducibility, Tool Validation, and Provenance

Forensic tool and method validation requires documented procedures, known datasets, and repeatable results before casework [2, 14, 18]. Chain-of-custody and provenance discipline demand that investigators show integrity of examined objects and distinguish observed from unavailable evidence [4, 11]. TrustEpisode’s contribution is not a higher detector F1; it is a single-owner revision contract, canonical replay boundary, and explicit terminal failure semantics that make reconstruction outcomes auditable under the same reproducibility expectations.

2.4 Incomplete Telemetry and Evidence-Loss Attribution

EDR and NDR fail differently; missing modality, parser blind spots, and collector health can erase or distort attack steps without implying that correlation “missed” an already-normalized event [17, 23]. Reconstruction contracts must therefore separate sensor observability, normalization/token mapping, formation membership, and offline attribution when diagnosing incomplete episodes.

2.5 Evidence-Grounded and Contained AI Investigation

Tool-using language-model systems interleave reasoning with external observations [25]. In security operations, CORTEX coordinates specialized agents over enterprise evidence, while Holmes packages network evidence into auditable anchors for an LLM investigator [5, 21]. These systems motivate evidence-grounded AI assistance, but grounding also needs an upstream contract: citation presence does not by itself establish faithfulness [7, 19], and agent execution requires architectural containment rather than prompt-only trust [24]. TRUSTEPISODE addresses the complementary boundary below the agent. It fixes which EDR/NDR objects exist, which producer owns them, how missingness is represented, and which immutable references a later narrative may cite. We do not claim a superior LLM or better autonomous triage.

2.6 Positioning Summary

Table 1 maps research units and objectives; it is not an effectiveness leaderboard. Our unit is *versioned evidence state*, not a score, suspicious subgraph, or narrative. The novelty claim is their contract composition: one membership writer, immutable successor/late-store semantics, source health separate from silence, one terminal

outcome per revision, stage-localized loss, and citable read-only records. Contract invariants and formation effectiveness are evaluated separately; no new detector or classifier is claimed.

2.7 Scope Distinction

TRUSTEPISODE does not claim a new graph detector, AI ranking effectiveness, attribution beyond observable EDR/NDR, enterprise impact estimation, or autonomous response. The evaluated contribution is a canonicalized deterministic forensic reconstruction and audit contract for sealed EDR/NDR evidence under late arrival and incomplete observability. Its AISec relevance is architectural: AI assistance is constrained to read and cite versioned case facts rather than silently redefining them.

3 TrustEpisode Reconstruction Contract

TRUSTEPISODE converts normalized EDR/NDR events and authenticated SourceCoverage into version-scoped forensic audit records that are deterministic over canonicalized, version-bound inputs. Episode Synthesis alone owns membership. The lifecycle is open episode $\rightarrow$ finalized revision $\rightarrow$ immutable late successor or closed lineage $\rightarrow$ beyond-horizon LateEvidenceRecord. Each revision yields exactly one terminal AssemblyOutcome (a canonical AuditRecord or an immutable AssemblyFailureRecord). Analysts and AI systems consume these records but cannot redefine membership, source health, probability support, or terminal status. Optional Feature Extractor / Scorer / Scope Resolver / Calibrator fields and sufficiency U remain design-only schema interfaces. Figure 1 shows these boundaries; ground truth is joined only offline.

3.1 Investigation and Threat Model

Assumptions. Raw pointers and the sealed evidence store are integrity-checkable; Health Monitor coverage messages have authenticated sources; attackers may induce missing events, delayed delivery, and duplicate retries; attackers cannot arbitrarily forge signed health state; runtime software may fault; the offline evaluator is isolated from online formation.

Non-assumptions. We do *not* assume complete telemetry, that collector silence equals downtime, that all attacks are observable, that ground truth is available online, that model scores are reliable, or that raw arrival order is fixed. Compromise of the Health Monitor identity, signing keys, or an external writer-authorization layer is outside scope; SHA-256 digests do not authenticate producers.

Downstream-agent adversary. An AI assistant may be mistaken, prompt-injected, or compromised: it may omit evidence, misquote a record, or recommend an unsafe action. It is outside the trusted computing base and has no capability to write canonical evidence objects. The contract does not make its narrative true; it makes any cited case claim checkable against an immutable revision and prevents the narrative from silently becoming the forensic record.

Investigator workflow. Investigators import sealed EDR/NDR, form revisions, and inspect each terminal audit outcome. They then check coverage and provenance, resolve late evidence, run the A-G waterfall for a missing step, replay the digests, and report observed, unavailable, and unknown separately.

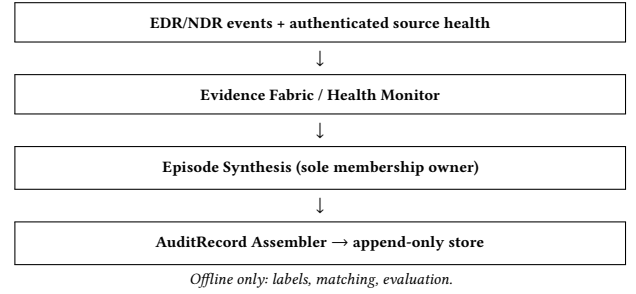


Figure 1: Label-blind online reconstruction and offline evaluation boundary.

3.2 Evidence and Reconstruction Model

The core evaluated objects are NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, AuditRecord, and AssemblyFailureRecord; optional design-only schema objects are FeatureVector, SupportDecision, nullable P , and structured U . Offline labels never enter online schemas. Table 2 assigns producers and forbidden behaviors for both.

3.3 Independent Evidence Health and Observability

Event silence is not an outage. “No detection” is not “unavailable.” SourceCoverage carries authenticated health, parser validity, and an observed source mask. Assemblers consume exact coverage references; they do not infer health from empty event windows.

3.4 Deterministic Episode Formation

Formation links events using process-causal, authenticated-session, endpoint-connection, and specific-entity relations under a 300 s maximum inter-event gap and 3600 s maximum episode span. Tie-breaks are deterministic under the frozen canonical ordering contract (relation precedence, relation count, Δt , start time, episode ID). Hub suppression, parent merge, and exact event-time ordering are part of the frozen policy. Canonical event-time ordering is a mandatory preprocessing step of the production formation path rather than an optional replay repair: raw delivery order is not robust, whereas re-canonicalized input reproduces byte-identical results. Formation never reads labels, receipts, or scenario metadata.

3.5 Immutable Revision Lifecycle

A watermark lag of 120 s advances the watermark behind observed ingest and finalizes open episodes whose event-time end is at or before that watermark. Late evidence inside a fixed 900 s revision horizon may create an immutable successor; beyond-horizon or closed-lineage evidence produces a LateEvidenceRecord. Earlier digests are never rewritten; deadlines are not extended by late arrivals.

3.6 Terminal Forensic Audit Contract

Open episodes finalize once their event-time end is at or before the advancing watermark (lag 120 s behind observed ingest). The assembler then applies a separate 120 s join deadline anchored at

Table 1: Related-work scope map (not an effectiveness comparison).

Research line	Primary unit	Primary objective	Relation to this work
Alert correlation [12]	alerts and scenario graphs	link multi-stage intrusion alerts	motivates temporal/causal linking; different output and evaluation target
Provenance APT analysis [6, 8–10, 20]	system/provenance graphs and suspicious subgraphs	detect, rank, or contextualize attack activity	complementary signal/context producers for a versioned evidence contract
Forensic reconstruction [1, 4, 16]	artifacts, time anchors, and timelines	reconstruct prior activity and document uncertainty	motivates chain-of-custody, late-evidence, and replay semantics
Evidence-grounded agents [5, 21, 25]	tool observations, citations, and narratives	orchestrate evidence and assist investigation	downstream consumers that need stable citable object identity
TrustEPISODE (this work)	versioned EDR/NDR episodes and terminal audit outcomes	make canonicalized reconstruction deterministic, append-only, loss-localized, and citable	evaluated contract; no detector or LLM-superiority claim

Table 2: Single-owner online object contract.

Object	Sole producer	Consumers	Forbidden behavior
NormalizedEvent	Evidence Fabric	Episode Synthesis	label, scenario, or health fields
SourceCoverage	Health Monitor	Scope Resolver; Sufficiency; Assembler	infer outage from event silence
EpisodeRevision / LateEvidenceRecord	Episode Synthesis	Feature; Sufficiency; Assembler / late store	rewrite a prior revision; admit beyond-horizon into membership
FeatureVector $[D, C]$	Feature Extractor	Stateless Scorer; Assembler	health/source mask; labels; scenario metadata
Nullable P	Calibrator (design)	Assembler	numeric unsupported output
Structured U	Sufficiency Evaluator	Assembler	scalarization that rewrites P
AssemblyOutcome	AuditRecord Assembler	Append-only store; replay	both success and failure; update/delete

Deterministic episode and revision lifecycle

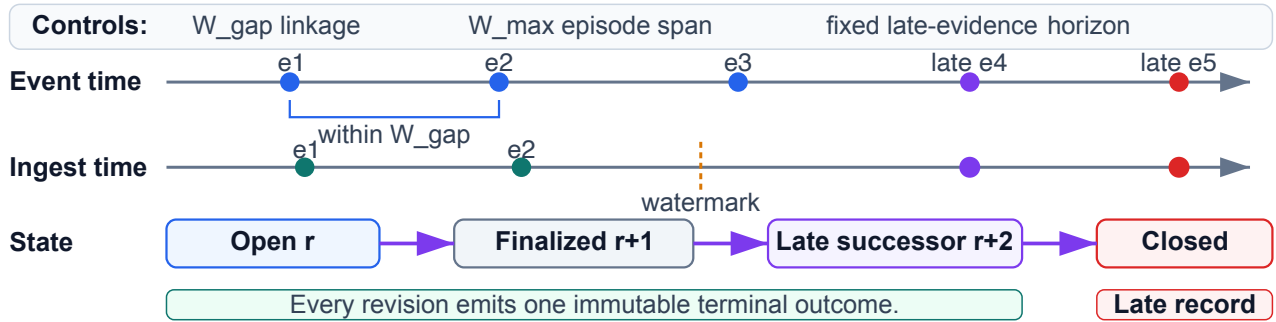


Figure 2: Episode revision lifecycle: open → finalized → late successor or closed; beyond-horizon evidence enters LateEvidenceRecord without rewriting earlier membership.

EpisodeRevision.watermark and inserts exactly one terminal outcome under $(episode_id, revision)$: AuditRecord or AssemblyFailureRecord (schema, key, duplicate, digest, version, missing-input, timeout). These are distinct lifecycle deadlines with different anchors. Success cannot follow terminal failure. References are canonically ordered; RFC 8785/JCS serialization and SHA-256 digests bind versions. The store is append-only.

3.7 Agent-Facing Evidence Boundary

Downstream analyst views and AI assistants are consumers, never producers, of canonical case facts. They receive a revision key, terminal AssemblyOutcome, ordered evidence references, SourceCoverage references, and the independent sufficiency profile. A generated explanation or recommended action is a noncanonical view: it cannot write EpisodeRevision, change P or U , or replace an AssemblyFailureRecord with success. If required inputs are missing, the only canonical output is the failure record; an agent is not permitted to fill the gap from conversational context. This boundary

makes a later narrative falsifiable against immutable source references without treating the language model as part of the forensic root of trust. Section 4.13 tests closed-schema, support-coupling, provenance, and digest guards. Section 4.14 then treats the model as an untrusted reader and scores exact values, abstention, and pointer citations; service-identity authorization remains a deployment assumption rather than an evaluated result.

Integrity is not writer authorization. The closed schema and unkeyed RFC 8785/SHA-256 digest validate object shape and detect an unsealed change; they do not authenticate the producer. A downstream principal with canonical write permission could change an allowed field and recompute the digest. The read-only boundary therefore requires two layers: externally enforced writer authorization, plus the evaluated schema/digest contract. AI7 is an expected-accept negative control that makes this dependency explicit rather than treating a hash as a signature.

Table 3: Evidence-loss taxonomy for incomplete reconstruction.

Class	Decision rule (summary)
A_execution_failure	Action receipt/predicate shows non-success
B_sensor_observability_failure	Receipt success but raw EDR/NDR lacks identifiable activity
C_parser_or_normalization_failure	Raw present; NormalizedEvent absent or rejected
D_token_mapping_failure	NormalizedEvent present under wrong/missing required token
E_formation_membership_failure	Correct token exists but not retained in matched episode
F_offline_attribution_failure	Episode retains evidence; matcher fails to credit token
G_unknown_or_incomplete_trace	Required supporting records missing for A-F

3.8 Evidence-Loss Taxonomy

When a required action token is absent from a reconstructed episode, diagnosis uses the following closed classes (Table 3):

Forbidden inference: treating episode-match $F1 = 1.0$ as exact-complete step recovery; treating missing tokens as formation failures without a waterfall. In operational cases without a trusted execution receipt or equivalent predicate record, the waterfall may terminate in G rather than distinguishing execution failure from downstream evidence loss. Synthetic A–G fixtures exercise the closed diagnostic taxonomy; they are not field validation.

3.9 Scoring and Probability Extension (Design Only)

Design-only and unevaluated here: a Stateless Scorer consumes only D and C and emits $z = f_{\theta}(D, C)$; a Scope Resolver separately consumes source availability / manifest scope and emits a SupportDecision governing whether P may be numeric. Source health must not influence z . Without supported scope and a fitted version-matched calibrator, P remains JSON null. Structured U cannot rewrite z or P . This submission reports no fitted coefficients, Brier/NLL, or held-out ranking.

3.10 Boundary

The *evaluated* contribution is deterministic reconstruction over canonicalized, version-bound inputs and replayable audit semantics, including evidence-loss localization and a deployed read-only case-materialization trace. Manifest-scoped probability is design-only. LLM generation quality, CTI enrichment, impact scoring, autonomous response, and OT/ICS generalization are excluded.

4 Evaluation

4.1 Evidence Classes and Cohorts

- **RQ1** reproducibility — sealed offline replay on all 60 primary malicious runs.
- **RQ2** lifecycle auditability — A1–A12 synthetic audit conformance covering seven failure codes + late/beyond-horizon cards.
- **RQ3** evidence-loss localization — A–G taxonomy conformance + sealed M2 waterfall.

- **RQ4** mechanism coverage — live M1–M7 + synthetic T1–T8, with M7 as a jointly out-of-contract temporal boundary family.
- **RQ5** baseline interpretation — sealed fair EB0–EB4 comparison on M1–M7.
- **RQ6** downstream boundary and reading — synthetic AI0–AI7 canonical-record validation plus two once-run 20-card local-model grounding audits.

Evidence classes used below are disjoint: (i) primary live empirical (M1–M7 formation cells); (ii) sealed offline replay on those primary seals; (iii) synthetic conformance (A1–A12, T1–T8, AB0–AB3, AI0–AI7, A–G fixtures); (iv) two once-run synthetic held-out LLM audits; (v) unit/integration boundary fixtures; (vi) supplemental partition artifacts (L1–L4, wall-clock) that are *not* mixed into primary estimators. Synthetic cards are never mixed into primary live effectiveness. Sealed Table 5 primary cells were not overwritten.

4.2 Primary Live M1–M7 Cohort

CALDERA v5.3.0 executes frozen Bash abilities on two isolated Ubuntu 22.04 targets with an EDR wrapper and inline Suricata NDR. Online schemas reject labels and receipts. The primary sealed cohort has 70 primary malicious formation runs (M1–M7 $\times$ two modes $\times$ five) plus 40 standalone benign execution controls. Because no fitted scorer or decision rule is evaluated, those controls are not false-positive trials and are not mixed into the malicious formation estimator. Fair sliding baselines and paired comparisons are computed on the M1–M7 set.

4.3 Ground Truth and Matching

Offline matching uses temporal IoU ≥ 0.10 , entity Jaccard ≥ 0.50 , overlapping required action tokens, and weight

$$w = 0.5A_{cov} + 0.3J_{entity} + 0.2IoU_{time}; \quad w \geq \theta_{match} = 0.50. \quad (1)$$

Qualified edges enter a maximum-weight one-to-one assignment between predicted episodes and truth groups. Metrics follow the sealed evaluator: *action coverage* is the mean best required-token fraction per truth group; *exact-complete recovery* is the share of groups with coverage = 1; *fragmentation/merge* count excess qualified edges per group or multi-group episodes; *contamination* is the benign-token share among retained actions on matched episodes (undefined without a denominator); *episode-match F1* is assignment precision/recall. Episode-match F1 measures whether an episode is matched to a truth group, not exact recovery of every required action. Online builders never read evaluation fields.

4.4 Fair Reconstruction Baselines

- **EB0_FIXED_BIN_300s** (legacy): fixed bins; boundary reference only.
- **EB1_SLIDING_TIME**: same $W_{gap} = 300$ s, $W_{max} = 3600$ s; time only.
- **EB2_SLIDING_ENTITY**: same lifecycle; entity overlap only.
- **EB3_RELATION_NO_ADVANCED_GUARDS**: relations on; hub/merge/late off.
- **EB4_TRUSTEPISODE**: full production policy.

Table 4: Primary M1–M7 scenario families (simplified). “Live EB1–EB4 delta?” reports whether fair builders differed on the sealed primary cohort for the intended advanced mechanism.

ID	Ordered activity (summary)	Expected EDR/NDR	Intended mechanism	Live EB1–EB4?
M1	recon → local action	process; DNS/flow	entity/time	identical
M2	remote session + SCP archive	process; SSH/SCP flows	session / file_transfer token	identical; token gap
M3	privilege / persistence steps	process; file	causal/entity	identical
M4	lateral-style dual-host steps	process; flows	session/endpoint	identical
M5	staged tooling / execution	process; file/net	entity/time	identical
M6	cleanup / cover tracks	process; file	entity/time	identical
M7	boundary actions at 0/300/3900 s	process; flows	joint temporal boundary	boundary failure

Primary comparison is EB1–EB4 on identical sealed streams, aliases, matching, and ground truth. Configuration digests are sealed in `results/derived/fair_baselines.json`.

4.5 Primary Formation Results and Fair Negative Result

Table 5 reports production-formation cells for the primary live M1–M7 cohort ($n = 5$ per cell). M7 is a deliberately out-of-envelope temporal boundary stress case (actions at 0/300/3900 s jointly violate $W_{gap} = 300$ s and $W_{max} = 3600$ s), so zero coverage and episode-match F1 are contract-conformant. It verifies expected rejection of a jointly out-of-envelope sequence and does not isolate W_{max} from W_{gap} ; Table 8 isolates those inclusive boundaries separately.

Relative to legacy EB0, the sealed cohort shows a descriptive mean coverage increase from 0.807 to 0.821 and exact-complete recovery from 0.671 to 0.714 (no EB4–EB0 paired CI is reported). Fixed bins cut multi-minute chains at arbitrary boundaries; sliding gap/sessionization preserves those chains under the same span cap. Across recoverable M1–M6 families, EB4 obtains mean action coverage of 0.958 and exact-complete recovery of 0.833; inclusion of the deliberately out-of-envelope M7 family reduces the aggregate to 0.821 and 0.714. Concurrent-benign M1–M6 episodes exhibit mean contamination 0.356: successful episode matching does not imply clean membership or reduced analyst workload, and contamination is an offline diagnostic rather than an online decision threshold.

The fair sliding baselines are not identical. EB4 matches EB2 on all action-coverage pairs (0/70/0 W/T/L). Against EB1, action coverage and exact-complete recovery have 0/69/1 W/T/L, mean $\Delta = -0.014$, and 95% CI $[-0.043, 0.000]$; episode-match F1 has mean $\Delta = -0.010$ and 95% CI $[-0.029, 0]$. The single loss is the concurrent-benign M7 run `run-ee27d58be2db032c44d0774d9358c4b1`: EB1 recovers its truth group while EB4 does not. We therefore report the EB4–EB1 result as *not statistically resolved*, not equivalence or relation-aware superiority. EB3 shows the same one-run action-coverage pattern.

What the fair negative result identifies. The fair result is informative about the benchmark: advanced relations are not the unique linkage path when EB4 matches EB2 on all 70 action-coverage pairs and EB1 on 69. A cohort can report strong episode-match F1 while barely exercising the mechanisms that distinguish a relation-aware contract from a sliding builder. This primary design mainly validates sliding sessionization; relation-specific guards need targeted mechanism-coverage evidence. Future BAS cohorts should register those requirements before claiming relation-aware superiority—session-only, endpoint-only, causal-only, organic hub over-merge,

parent merge, within-horizon late successor, beyond-horizon late evidence, and isolated W_{gap}/W_{max} boundaries. Synthetic cards show conformance, not organic live effectiveness. M7’s zero recovery remains the expected jointly out-of-envelope outcome.

4.6 Deployed Case-Materialization Trace

We additionally recorded a read-only trace from the deployed investigation surface on 2026-07-23. This trace is artifact-level integration evidence, not another experimental sample and not part of any estimator above. Figure 3(a) shows the selected laboratory case: an investigation-control summary over three normalized events—one Suricata NDR event and two Wazuh events—with source-separated evidence cards and an ATT&CK support path for T1041 and T1018. Figure 3(b) shows the corresponding entity-relationship graph, which links the external source, the NDR and Wazuh evidence, and the target host from the same `graph_nodes/graph_edges` records. The interface’s 98% operational confidence and `isolate_or_investigate` recommendation belong to the downstream case engine; they are *not* TRUSTEPISODE’s nullable calibrated P and are not used as evidence of model accuracy. The trace supports only the narrower claim that source-separated evidence and technique support survive into a human-reviewable case view.

4.7 Inclusive Temporal Boundary Fixtures

Unit fixtures on the production candidate function confirm inclusive equality at the frozen contracts (Table 8). For W_{gap} , 299 and 300 s are eligible while 301 s is not. For W_{max} , 3599 and 3600 s are eligible while 3601 and 3900 s are not. The 3900 s case matches the sealed M7 schedule offset and is *not* a substitute for live M7 formation metrics. The companion seals the complete boundary result.

4.8 Sixty-Run Determinism Matrix

On all 60 sealed malicious runs (M1–M6 $\times$ attack-only/concurrent-benign $\times$ five), D1 canonical ordered replay (10 $\times$ per run; 600 executions) is byte-identical (60/60). D2 separates raw delivery-order robustness (0/60) from re-canonicalized equality (60/60): we do *not* claim raw-order robustness. D3 ingest-delay grids (including inside, at, and beyond the 900 s revision horizon) pass 60/60; D4 duplicate/idempotent/conflict cases pass 60/60. D5 documents the single-writer architectural assumption rather than reporting an executed concurrency test: multi-run parallelism is not same-run concurrent mutation, and we claim neither lock-free nor multi-writer determinism.

Table 5: Primary live M1–M7 cells under production formation (mean $\pm$ sample SD; $n = 5$).

Family	Mode	Coverage	Frag.	Merge	Contam.	Match F1
M1	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M1	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.300 $\pm$ 0.112	1.000 $\pm$ 0.000
M2	attack-only	0.750 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M2	concurrent-benign	0.750 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.350 $\pm$ 0.091	1.000 $\pm$ 0.000
M3	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M3	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.350 $\pm$ 0.091	1.000 $\pm$ 0.000
M4	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M4	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.367 $\pm$ 0.217	1.000 $\pm$ 0.000
M5	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M5	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.433 $\pm$ 0.091	1.000 $\pm$ 0.000
M6	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M6	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.333 $\pm$ 0.257	1.000 $\pm$ 0.000
M7	attack-only	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	undef.	0.000 $\pm$ 0.000
M7	concurrent-benign	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	undef.	0.000 $\pm$ 0.000

Table 6: Fair sliding-gap baselines versus legacy fixed bins on the sealed 70-run M1–M7 malicious cohort (mean $\pm$ sample SD). Primary comparisons share $W_{gap} = 300$ s and $W_{max} = 3600$ s.

Builder	Role	Coverage	Exact-comp.	Frag.	Contam.	Match F1
EB0_FIXED_BIN_300s	legacy	0.807 $\pm$ 0.348	0.671 $\pm$ 0.473	0.071 $\pm$ 0.259	0.178 $\pm$ 0.208	0.814 $\pm$ 0.353
EB1_SLIDING_TIME	primary	0.836 $\pm$ 0.335	0.729 $\pm$ 0.448	0.000 $\pm$ 0.000	0.175 $\pm$ 0.207	0.867 $\pm$ 0.338
EB2_SLIDING_ENTITY	primary	0.821 $\pm$ 0.349	0.714 $\pm$ 0.455	0.000 $\pm$ 0.000	0.178 $\pm$ 0.208	0.857 $\pm$ 0.352
EB3_RELATION_NO_ADVANCED_GUARDS	primary	0.836 $\pm$ 0.335	0.729 $\pm$ 0.448	0.000 $\pm$ 0.000	0.175 $\pm$ 0.207	0.863 $\pm$ 0.341
EB4_TRUSTEPISEDE	primary	0.821 $\pm$ 0.349	0.714 $\pm$ 0.455	0.000 $\pm$ 0.000	0.178 $\pm$ 0.208	0.857 $\pm$ 0.352

Table 7: Paired differences EB4–EB1_SLIDING_TIME (contamination/fragmentation flipped so positive favors EB4). Bootstrap: 2000 run-cluster resamples, seed 20260721. Zero deltas reported as no observed difference.

Metric	Mean Δ	Median Δ	95% CI	W/T/L	Interpretation
action_coverage	−0.014	0.000	[−0.043, 0.000]	0/69/1	not resolved
exact_complete_recovery	−0.014	0.000	[−0.043, 0.000]	0/69/1	not resolved
fragmentation	0.000	0.000	[0.000, 0.000]	0/70/0	no observed difference
contamination	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference
episode_match_f1	−0.010	0.000	[−0.029, 0.000]	0/69/1	not resolved

Table 8: Inclusive W_{gap}/W_{max} boundary fixtures (unit; all pass).

Dimension	Seconds	Expected	Observed
W_{gap}	299	eligible	eligible
W_{gap}	300	eligible	eligible
W_{gap}	301	ineligible	ineligible
W_{max}	3599	eligible	eligible
W_{max}	3600	eligible	eligible
W_{max}	3601	ineligible	ineligible
W_{max}	3900	ineligible	ineligible

4.9 Audit Failure Conformance (A1–A12)

RQ2 is supported by a matrix covering all seven predefined AssemblyFailureRecord codes (7/7) with expected terminal outcomes, plus precedence, exclusivity, idempotent success retry, and append-only rejection (18/18 fixtures pass). This is synthetic forensic audit conformance, not field effectiveness.

4.10 Evidence-Loss Taxonomy Validation

The taxonomy is conformance-tested across A–G and instantiated on one sealed live case, M2 (8/8 fixture rows; 1 live). Synthetic passes do not imply field-validated coverage for every class.

4.11 M2 Evidence Waterfall

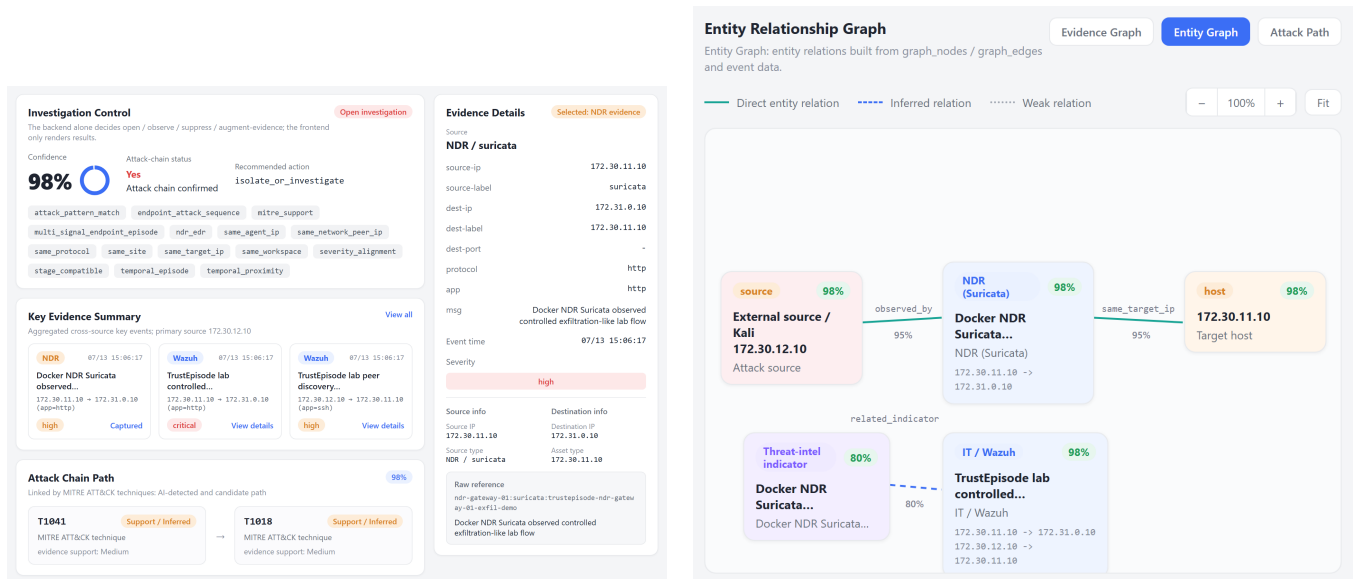
Final class: D_token_mapping failure. Action coverage 0.75 with episode-match F1 1.0 is therefore not evidence that Episode Synthesis dropped an already-normalized file_transfer event. Primary sealed results are unchanged; any mapper fix requires a new version digest and a post-analysis cohort.

4.12 Contract Ablation Baselines (AB0–AB3)

AB0–AB3 are reference implementations for an isolation study: which contract property prevents which forensic failure mode. They are reference implementations for isolation, not product comparisons.

4.13 Agent-Facing Boundary Conformance

AI0 accepts an unchanged canonical AuditRecord. AI1–AI4 inject a generated fact, leak a label, force an unsupported 0.98 probability, or remove provenance; the closed schema rejects each. AI5–AI6 change a revision or evidence digest without resealing; digest verification rejects both. AI7 changes an allowed D field and reseals with the unkeyed RFC 8785/SHA-256 digest; validation accepts it. Thus 8/8 cards match their specifications. Integrity verification detects



(a) Selected case: investigation control, source-separated key evidence, and the ATT&CK support path, with the NDR evidence detail. (b) Entity-relationship graph linking the external source, NDR/Wazuh evidence, and the target host.

Figure 3: Read-only deployed case trace from the isolated TrustEpisode laboratory (light-theme, English interface). The panels are cropped to remove account-identifying chrome; they demonstrate end-to-end evidence materialization, not calibrated probability or LLM effectiveness. The displayed operational confidence and recommendation are downstream UI fields, not TrustEpisode probability outputs and not evidence of model effectiveness.

Table 9: Determinism matrix on all 60 sealed malicious runs (D1–D5).

Test	Runs	Variants	Expected invariant	Pass	Fail	Interpretation
D1 canonical replay	60	10×/run	byte-identical digests	60	0	canonical ordered input
D2 raw delivery order	60	6 perms	equals baseline without reorder	0	60	not order-robust
D2 re-canonicalized	60	6 perms	equals baseline after sort	60	0	canonicalization required
D3 ingest delay	60	horizon grid	successor/late invariants	60	0	prior digest preserved
D4 duplicates	60	conflict/idempotent	reject or no-op	60	0	fail-closed conflicts
D5 single-writer	architecture	ownership	no same-run multi-writer	N/A	N/A	architectural constraint; not experimentally tested

Table 10: A1–A12 audit conformance fixtures (synthetic; covers all seven AssemblyFailureRecord codes).

Fixture	Expected	Observed	Term.	Prior chg?	Result
A1_SUCCESS	success	success	1	no	pass
A2_SCHEMA_INVALID	schema_invalid	schema_invalid	1	no	pass
A3_KEY_MISMATCH	key_mismatch	key_mismatch	1	no	pass
A4_CONFLICTING_DUPLICATE	conflicting_duplicate	conflicting_duplicate	1	no	pass
A5_DIGEST_MISMATCH	digest_mismatch	digest_mismatch	1	no	pass
A6_VERSION_MISMATCH	version_mismatch	version_mismatch	1	no	pass
A7_MISSING_INPUT	missing_input	missing_input	1	no	pass
A8_ASSEMBLY_TIMEOUT	assembly_timeout	assembly_timeout	1	no	pass
A9_FAILURE_PRECEDENCE_SCHEMA	schema_invalid	schema_invalid	1	no	pass
A9_FAILURE_PRECEDENCE_KEY	key_mismatch	key_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_DUPLICATE	conflicting_duplicate	conflicting_duplicate	1	no	pass
A9_FAILURE_PRECEDENCE_DIGEST	digest_mismatch	digest_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_VERSION	version_mismatch	version_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_MISSING	missing_input	missing_input	1	no	pass
A9_FAILURE_PRECEDENCE_TIMEOUT	assembly_timeout	assembly_timeout	1	no	pass
A10_TERMINAL_EXCLUSIVITY	rejected	rejected	1	no	pass
A11_IDEMPOTENT_SUCCESS_RETRY	idempotent	idempotent	1	no	pass
A12_APPEND_ONLY_STORE	both_rejected	both_rejected	1	no	pass

invalid or unsealed mutations; write prevention depends on external authorization. These cards evaluate canonical-record boundary conformance (property A), not LLM factuality.

4.14 Held-Out Local-Model Grounding Audits

After an unscored smoke test, we froze the prompt, schema, four re-digested AuditRecords, and 20 held-out tasks, then ran qwen2.5:7b

Table 11: A–G evidence-loss taxonomy conformance (synthetic A–G plus one sealed live M2 as D).

Expected class	Observed class	Source	Result
A_execution_failure	A_execution_failure	synthetic	pass
B_sensor_observability_failure	B_sensor_observability_failure	synthetic	pass
C_parser_or_normalization_failure	C_parser_or_normalization_failure	synthetic	pass
D_token_mapping_failure	D_token_mapping_failure	synthetic	pass
E_formation_membership_failure	E_formation_membership_failure	synthetic	pass
F_offline_attribution_failure	F_offline_attribution_failure	synthetic	pass
G_unknown_or_incomplete_trace	G_unknown_or_incomplete_trace	synthetic	pass
D_token_mapping_failure	D_token_mapping_failure	live M2	pass

Table 12: M2 evidence-loss waterfall (10 sealed runs).

Stage	Expected	Observed	Status
CALDERA receipt (linux.scp_archive)	10	10	retained
Raw EDR/NDR observability (scp mentions)	10	10	retained
NormalizedEvent as file_transfer	10	0	drop
Normalized as remote_session instead	10	10	remapped
Episode retains file_transfer	10	0	absent token
Episode retains remote_session	10	10	retained

once (temperature 0, seed 20260723). After observing Qwen but before querying deployment-default gemma3:1b, we locked the same protocol and ran Gemma once. Table 15 is a synthetic, two-model, same-card, once-run transfer audit of untrusted-reader behavior (property B): extraction, abstention, and citation. Qwen yields 9/20 fully exact responses; Gemma yields 0/20. The cross-model failures show why the model must remain outside the forensic root of trust and why typed evidence identity is needed for post-hoc verification.

4.15 Mechanism Conformance Cards

T1–T8 are synthetic. T2/T3 were redesigned so that specific_entity is disabled and session/endpoint becomes the sole legal relation; ablation then fragments. Passes are conformance, not live effectiveness. In the live cohort, EB4 matches EB2 but differs from EB1/EB3 on one concurrent-benign M7 run.

4.16 Supplemental Partition (Not Primary)

Artifacts below live under a separate supplemental_mechanism partition. They do *not* enter Table 5 or the 70-run fair baseline estimator. Evidence classes are stated per row.

- **L1 (live base + offline ablation):** three sealed live M3 SSH bases. With only session linking enabled, the offline ablation links each base; disabling it fragments all three. This is not organic session-only BAS stress.
- **L2 (offline ablation on L1 seals):** on the same seals, endpoint-only linking is denser than the endpoint-disabled ablation (3/3). This is an offline mechanism check.
- **L3 (offline ablation + controlled inject):** organic hub load is *not* observed (hub-on vs hub-off identical, 0/3 organic). Controlled hub-load inject on the live sealed bases (cutoff+1 peers) makes hub-on vs hub-off diverge (3/3). Controlled inject $\neq$ organic BAS hub stress.
- **L4 late evidence (two distinct classes):** (i) *controlled late inject* on live seals yields successors and beyond_revision_horizon records (3/3). (ii) *A wall-clock hold/release* uses real delays: within-horizon 180 s holds seal and show multi-revision successors (2/2 pass); beyond-horizon attempts

(930 s hold-release; 1080 s quiet wait) *seal* but do *not* emit LateEvidenceRecords under organic telemetry (2 sealed, 0/2 LateEvidence; new membership formed instead). Beyond-horizon localization is therefore supported by controlled inject, not by wall-clock beyond runs in this lab.

4.17 Artifact Reproducibility

The anonymous companion archive fails closed unless Python imports its packaged reference runtime, executes all 38 implementation tests, and re-runs A1–A12, A–G, AB0–AB3, AI0–AI7, T1–T8, and the temporal boundary fixtures. It regenerates tables from sealed JSON and checks the reported D1–D5, fair-baseline, and two local-model summaries; raw/derived equality is checked for both model audits. It does not re-run live collection, cohort re-synthesis that requires the omitted raw lab bundles, or Ollama. Every clean-room command executes inside the extracted ZIP; the adjacent report records the results. Digests are refreshed with each packing.

Ethical scope. All traffic and workloads were confined to isolated, author-controlled laboratory systems. The study used no production users, personal data, human subjects, or third-party targets; screenshots were cropped to exclude account-identifying chrome.

4.18 Limitations

External validity. Two Ubuntu targets, a non-commercial EDR wrapper, and controlled Suricata/Wazuh-style telemetry; no Windows or commercial EDR and no production base rates. **Statistical and effectiveness limits.** $n=5$ per cell; benign controls are not false-positive trials; fair sliding results largely overlap, with one concurrent-benign M7 loss of EB4 versus EB1/EB3; concurrent-benign contamination remains nontrivial. **Mechanism coverage.** Organic hub stress is absent; wall-clock beyond-horizon LateEvidenceRecord is not observed (2 sealed, 0/2); M7 jointly stresses W_{gap} and W_{max} ; session/endpoint evidence is partly offline ablation. **Unevaluated extensions.** Fitted scoring/calibration, partial-telemetry/outage replay, writer authorization, and general LLM factuality remain outside the claims; raw arrival order requires re-canonicalization; M2 token mapping is unrepaired in the primary cohort.

Analysis status. Completed: 70 malicious + 40 benign controls; fair EB0–EB4; T1–T8; M2; D1–D5; A1–A12; A–G; AB0–AB3; AI0–AI7; Qwen + locked Gemma; inclusive boundary fixtures. Supplemental: L1–L4. Not executed: organic hub-stressed BAS; fitted scorer / RP / PO1.

Table 13: Contract ablation / reference-implementation comparison (synthetic isolation study; not SOTA product baselines).

Property	AB0 Mutable	AB1 Noncan.	AB2 Silent	TrustEpisode	Evidence
prior_history_immutable_under_late	no	no	yes	yes	synthetic conformance
byte_identical_replay_under_ref_reorder	yes	no	yes	yes	synthetic conformance
explicit_terminal_failure_on_missing_input	no	no	no	yes	synthetic conformance
append_only_terminal_store	no	yes	yes	yes	synthetic conformance

Table 14: Downstream-agent canonical-record boundary cards (synthetic conformance only; not LLM accuracy or deployed authorization).

Card	Operation	Guard	Expected	Observed	Result
AI0	consume unchanged canonical AuditRecord	closed schema plus canonical digest	accept	accept	pass
AI1	add agent_asserted_fact to canonical record	additionalProperties false	schema reject	schema reject	pass
AI2	add malicious_label to online record	additionalProperties false	schema reject	schema reject	pass
AI3	replace out-of-support null P with 0.98	support-probability coupling	schema reject	schema reject	pass
AI4	remove ordered evidence references	required provenance field	schema reject	schema reject	pass
AI5	swap revision digest without resealing record	canonical digest binding	digest reject	digest reject	pass
AI6	swap evidence content digest without resealing record	canonical digest binding	digest reject	digest reject	pass
AI7	replace allowed D and recompute unkeyed digest	external writer authorization required	accept	accept	pass

Table 15: Held-out AuditRecord pilots. Gemma was locked after the Qwen result but before model download. Exact requires answer, abstention flag, and citation set to match.

Model	JSON	Answer	Abst.	Cite	Exact	Bad ptr.
Qwen2.5-7B	20/20	15/20	14/20	15/20	9/20	1
Gemma3-1B	15/20	7/20	13/20	0/20	0/20	18

5 Conclusion

TRUSTEPISODE contributes a canonicalized, single-writer evidence contract for AI-assisted EDR/NDR investigation. It preserves immutable revision history, emits append-only terminal outcomes, localizes evidence loss with the A–G taxonomy, and exposes a typed read-only case boundary to downstream AI. Formation superiority was not established against fair sliding builders: EB4 matches EB2 on all 70 action-coverage pairs and trails EB1/EB3 on one concurrent-benign M7 run ($\Delta = -0.014$, 95% CI $[-0.043, 0.000]$; not statistically resolved). That negative result shows that the live cohort mainly exercises sliding sessionization and motivates explicit mechanism-coverage testing. Canonical replay succeeds over canonicalized inputs; raw arrival order requires re-canonicalization. Downstream mutation cards separate integrity verification from writer authentication; once-run Qwen and Gemma audits (9/20 and 0/20) demonstrate inspectability rather than general model correctness. TRUSTEPISODE does not make an untrusted model correct. It establishes a falsifiable evidence substrate in which canonical case facts, missingness, revision history, and model citations can be independently inspected.

A Frozen Submission Contract

Machine and audit contract. The normative companion uses closed JSON Schema Draft 2020-12 objects [22] for NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, FeatureVector, SupportDecision, AuditRecord, and AssemblyFailureRecord. FeatureVector cannot consume health state. SupportDecision resolves detector availability, support, group key, and eligibility; zero eligible detectors forces out-of-support, false eligibility, and JSON null probability. For each revision, the Assembler inserts exactly one terminal AuditRecord after a complete version-matched join

or one AssemblyFailureRecord after schema, key, duplicate, digest, version, missing-input, or timeout failure. Failure cannot later be replaced by success. Online objects reject labels, scenario/card IDs, receipts, and attack windows. Canonical bytes follow RFC 8785 [13]; application-ordered reference arrays are sorted before SHA-256, and every schema, parser, formation, detector, model, support, sufficiency, assembly, and manifest version binds both identifier and content digest.

Episode and lifecycle constants. Frozen values are $W_{gap} = 300$ s, $W_{max} = 3600$ s, watermark lag 120 s, revision horizon 900 s, and closed-lineage retention 86400 s. Candidate relations rank causal/session/endpoint/ entity as (4, 3, 2, 1) and sort by relation rank, number of satisfied classes, time distance, start time, then episode ID. A hub has at least 25 relations in 600 s; merge admits at most three causally/session/endpoint-connected parents and 500 events/200 entities. Events sort lexicographically by event time, ingest time, source-instance bytes, and event-ID bytes. First finalization fixes the revision deadline. A qualifying event before it creates an immutable successor without extending the horizon; a later related event creates only a LateEvidenceRecord. Replay equality covers states, successors, late records, ordered references, and digests.

Evaluation and artifact boundary. The achieved primary cohort contains five valid runs per M1–M7 card/mode and five standalone runs per B1–B8 card. The B1–B8 runs are execution controls, not false-positive trials because no fitted decision rule is evaluated. Offline evaluation freezes malicious-coverage 0.50, match weight 0.50, temporal IoU 0.10, and entity Jaccard 0.50; contamination 0.25 is only a positive/mixed offline-label diagnostic and does not reject a qualified match. Paired comparisons use a 2000-resample run-cluster bootstrap with seed 20260721. Fitted calibration, RP0–RP7 partial-telemetry replay, physical outage, and full cost benchmarking remain unexecuted and are not claimed. The companion contains the full schemas, thresholds, cards, validators, sealed result copies, and claim matrix. Its filename is TrustEpisode_reproducibility_companion_v1.zip; concatenating the following lines gives its SHA-256:

```
cc4d451b50f975fdd7ab07c263a8ce20
44ab771126808e485faac4033852a730
```

For anonymous review, the companion contents are browsable at <https://anonymous.4open.science/r/trustepisode-artifact-9E59/>; the authoritative digest above is for the packaged archive, which reviewers should verify before extraction.

Table 16: Targeted synthetic mechanism cards (conformance only; not mixed into primary effectiveness).

Card	Result	Expected	Observed
T1-PROCESS-CAUSAL	pass	causal-off fragments	full 1; ablated 2
T2-AUTH-SESSION	pass	session-off fragments	full 1; ablated 2
T3-ENDPOINT-MAPPING	pass	endpoint-off fragments	full 1; ablated 2
T4-HUB-GUARD	pass	hub guard changes membership	full 4; no-hub 1
T5-PARENT-MERGE	pass	merge creates parents	full 3; ablated 0
T6-LATE-SUCCESSOR	pass	late successor preserved	full rev. 4; ablated 3
T7-BEYOND-HORIZON	pass	beyond horizon → late record	late record 1
T8-TIE-BREAK	pass	canonical replay equal	sorted equal; raw differs

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